

Ultrasound and Doppler of the Child's Liver.

In the last chapter we took a needle to the liver. In this chapter we look at the liver without touching it. Ultrasound is the first picture we take of almost every child with a liver problem. It is cheap, it can be done at the bedside, it needs no radiation, and in most children it needs no sedation. For a jaundiced infant, a child with a big spleen, or a child whose liver enzymes rise after a transplant, it is usually the first imaging test, and often the only one.

But ultrasound has two weaknesses. The first is that it depends on the person holding the probe. The same baby can have a normal scan in one hospital and an abnormal scan in another. The second weakness is more specific to our field. Most of the rules that we use to read a liver Doppler study were worked out in adults. Some of them work in children. Some of them do not work at all. And a few of them point in exactly the wrong direction.

So this chapter asks five questions. First, what should a complete ultrasound report of a child's liver contain? Second, how good is ultrasound at finding biliary atresia, and what should we do when the scan is normal? Third, which adult Doppler rules survive in children? Fourth, what can Doppler tell us about prognosis and about portal hypertension? And fifth, how do we read the Doppler study of a liver graft after transplant?

A short note on the sources. For the cholestatic infant, the main document is the joint guideline of the North American and European paediatric societies, NASPGHAN and ESPGHAN, written by Fawaz and colleagues in 2017. For portal hypertension in children, we will use the Baveno six paediatric consensus of 2016, written by Shneider and colleagues, and for adults the Baveno seven consensus of 2022. For vascular disease of the liver, we will use the American guidance from AASLD in 2020 and the European guideline from EASL in 2016. For contrast ultrasound in children, we will use the European ultrasound federation, EFSUMB, in 2017. Most of the numbers in this chapter, however, come not from guidelines but from single studies, and we will name them as we go.

A note on words. The resistive index of an artery is a number between zero and one. It compares the speed of blood in systole with the speed in diastole. A high resistive index means that little blood flows in diastole, so the bed downstream is resisting the flow. Flow in the portal vein towards the liver is called hepatopetal, which is normal. Flow away from the liver is called hepatofugal, which is abnormal. And peak systolic velocity is simply the fastest speed of the blood in systole, measured in centimetres per second.

Part one. What the report must contain.

A good report is not a description. It is a checklist. There are seven structures that should appear in every report of a child's liver, and for each one there is a measurement, a normal value in children, and a reason why it matters.

The first is the gallbladder. The child should fast for at least four hours before the scan, because a gallbladder that has just emptied after a feed looks small and abnormal. The radiologist should measure its length, and look at its wall, its lining and its outline. In a normal infant, the gallbladder is at least one point nine centimetres long, and it has a smooth, bright lining. Some centres also look at whether it contracts after a feed. Why does this matter? Because a small, irregular gallbladder without a bright lining is one of the strongest signs of biliary atresia.

The second is the porta hepatis, the place where the bile duct and the vessels enter the liver. Here the radiologist looks for a band of bright tissue just in front of the right portal vein. This is called the triangular cord. Normally it is four millimetres thick or less. The radiologist also looks for a cyst at the porta. Why does this matter? A thick cord suggests biliary atresia, and a cyst raises the question of cystic biliary atresia versus a choledochal cyst.

The third is the hepatic artery. The radiologist measures the diameter of the right hepatic artery, which is normally less than one point five millimetres in an infant, and the resistive index, which is about zero point six in healthy children, with a range of about zero point one either side. The resistive index falls a little as children grow. The radiologist also looks for colour flow along the surface of the liver, just under the capsule, which is called subcapsular flow. A large artery and subcapsular flow are both seen in biliary atresia.

The fourth is the portal vein. The radiologist reports its diameter, the direction of flow, the speed, and whether the flow pulses with the heart. Normally the flow goes into the liver, and the maximum speed is above sixteen centimetres per second. This matters for portal hypertension, and before a transplant, because a small portal vein changes how the surgeon plans the operation.

The fifth is the hepatic veins. In adults, a normal hepatic vein has a wave with three phases, called a triphasic wave. But, as we will see, only about four in ten healthy children have a triphasic wave in all three hepatic veins. So in children, the most useful comparison is with the same child's earlier scan.

The sixth is the spleen. Its length is measured on a coronal view, from the top to the tip, through the hilum. The normal length depends on age. The upper limit is about six centimetres at three months and about thirteen centimetres in boys over fifteen years.

The seventh is the body plan and the collaterals. Is the child's anatomy normal, or is there a heterotaxy, with several small spleens, a portal vein that runs in front of the duodenum, or an interrupted inferior vena cava? And are there collateral vessels, such as a reopened umbilical vein or a shunt between the spleen and the kidney? Normally none of these are present. The anatomy matters because it points to the splenic malformation form of biliary atresia, and because the surgeon needs to know. The collaterals matter because they are a direct sign of portal hypertension.

If one of these seven items is missing from the report, ask for it. The radiologist will usually be glad to add it.

Part two. Seven signs of biliary atresia.

Biliary atresia is the most urgent diagnosis in infant liver disease, because the result of the Kasai operation depends on how early it is done. Ultrasound cannot make the final diagnosis. That is the job of the cholangiogram. But ultrasound is the first test, and it can raise or lower our suspicion.

There are seven ultrasound signs of biliary atresia. Let us go through them, with their sensitivity and specificity. The best summary comes from a meta-analysis published by Zhou and colleagues in 2016.

The first sign is the triangular cord, the band of bright tissue in front of the right portal vein, more than four millimetres thick on a longitudinal scan. In the Zhou meta-analysis of twenty studies, its sensitivity was seventy-four percent and its specificity ninety-seven percent.

The second sign is an abnormal gallbladder. This means a length under one point nine centimetres, an irregular outline, or a wall or lining that cannot be clearly seen. In nineteen studies, its sensitivity was eighty-five percent and its specificity ninety-two percent.

The third sign is a gallbladder that cannot be seen at all after a four-hour fast. Its specificity was ninety-nine percent in ten studies. Its sensitivity is lower, because many infants with atresia still have a small gallbladder.

The fourth sign is subcapsular flow, colour flow along the surface of the liver. In a study of sixty-four infants by Lee and colleagues in 2009, it had a sensitivity of one hundred percent and a specificity of eighty-six percent.

The fifth sign is a right hepatic artery wider than one point five millimetres. In sixty-eight infants studied by Kim and colleagues in 2007, its sensitivity was ninety-two percent and its specificity eighty-seven percent.

The sixth sign is a common bile duct that cannot be seen with a high-frequency probe. In ninety infants studied by Humphrey and Stringer in Leeds in 2007, this had a sensitivity of ninety-three percent and a specificity of ninety-two percent.

The seventh sign is a cyst at the porta hepatis, together with an abnormal gallbladder. This points to cystic biliary atresia. In a meta-analysis by Yang and colleagues in 2025, it was quite specific, at eighty-four percent, but it was seen in only seventeen percent of infants with atresia.

Notice a pattern. The signs that describe shape, the triangular cord and the abnormal gallbladder, are very specific. When they are present, atresia is likely. The signs that describe blood flow, the subcapsular flow and the large hepatic artery, are the most sensitive. They catch more infants, but they also raise more false alarms.

Part three. The triangular cord and the ghost gallbladder.

Let us look more closely at the two most famous signs.

The triangular cord is specific everywhere, and sensitive nowhere. Look at the range of sensitivity across studies. A meta-analysis by Yoon and colleagues in 2017, with seventeen studies, found eighty-five percent. A study from Daegu in Korea, by Lee and colleagues in 2003, found eighty percent. Tan Kendrick and colleagues in Singapore, in 2003, found seventy-seven percent. The Zhou meta-analysis found seventy-four percent. Humphrey in Leeds found seventy-three percent. Lee in Seoul, in 2009, found sixty-two percent. A network meta-analysis by Zhang and colleagues in 2024 found fifty-eight percent. And a study from Chandigarh in India, published by Mittal and colleagues in 2011, found only twenty-three percent.

The specificity, on the other hand, was between ninety-seven and one hundred percent in every series except the network meta-analysis, where it was ninety-three percent.

Why is the sensitivity so variable? Part of the answer is the cut-off. Some centres call the cord positive at three millimetres and others at four. Part of the answer is age. The cord is fibrous tissue at the hilum, and in a very young infant this fibrosis may not have formed yet. The Chandigarh infants were all under ninety days old.

So the practical rule is simple. A positive triangular cord is something you can act on. A negative triangular cord changes nothing.

Now the gallbladder. In 2003, Tan Kendrick and colleagues in Singapore described what they called the gallbladder ghost triad. It has three parts. First, a length under one point nine centimetres. Second, no smooth, bright lining. Third,

an irregular or lobulated outline. Thirty of thirty-one infants with biliary atresia had this triad, and there were no false positives among one hundred and eighty-six infants without atresia.

The one missed infant is instructive. At six weeks, the gallbladder looked normal. Two weeks later, at eight weeks, it had become a ghost. The gallbladder can lag behind the disease.

A study by Farrant and colleagues in 2000, in three hundred and forty-six infants, gives the other side of the picture. A gallbladder with a normal shape and a normal wall had a negative predictive value of ninety-seven point three percent. So a normal gallbladder makes atresia much less likely, but not impossible.

There is one more clue. In one series, a thick gallbladder wall was seen in forty-six infants without atresia and in none with atresia. So a thick wall argues for neonatal hepatitis, not for atresia.

What happens when we combine signs? In the Zhou meta-analysis, combining the triangular cord with the gallbladder findings raised the sensitivity to ninety-five percent, while the specificity fell to eighty-nine percent. For a screening scan, that is the right trade. We would rather have a few false alarms than miss an infant who needs a Kasai operation. And the Doppler signs, subcapsular flow and a large hepatic artery, are best used as reasons to look harder, not as reasons to decide.

Part four. A normal scan is not a normal child.

The most important sentence about ultrasound in the cholestatic infant comes from the NASPGHAN and ESPGHAN guideline of 2017. It is a strong recommendation with high-quality evidence. A normal ultrasound does not rule out biliary atresia.

Why can the scan be normal? There are three reasons.

The first is biology. In type one atresia and in cystic atresia, the gallbladder may still be present and may look quite normal. And the triangular cord needs a fibrous plate at the hilum, which is often not yet formed early in the disease.

The second is technique. A baby who has just fed will have a contracted gallbladder that can look exactly like a ghost. So the fast matters. And the triangular cord and the small duct at the hilum can only be seen with a high-frequency linear probe, of about twelve to fifteen megahertz. A standard abdominal probe will miss them.

The third is the cut-off. As we saw, some centres use three millimetres for the cord and others use four, and this explains much of the difference between studies.

What should we do? The guideline says that ultrasound is the first imaging test in every cholestatic infant, and that it should be done after fasting. It reliably shows a choledochal cyst and gallstones. But if the stools stay pale, or the GGT keeps rising, the next step is a biopsy or a cholangiogram, whatever the scan showed. And if the first scan was normal and the stools are still pale, scan again.

There is one exception to the rule about fasting. An infant with hypopituitarism, which is one cause of neonatal cholestasis, can become dangerously hypoglycaemic. This infant should be scanned without a long fast.

And finally, when you scan an infant with suspected atresia, look for the splenic malformation syndrome as well. Several small spleens, a portal vein running in front of the duodenum, an interrupted inferior vena cava and a liver lying in the middle of the abdomen all point to this form of atresia. If you find them, ask for an echocardiogram, because heart defects are common, and tell the surgeon.

Part five. Adult Doppler rules in children.

Most of what we know about liver Doppler comes from adults. Let us go through seven common adult rules and see what the paediatric data show.

The first rule says that a hepatic artery resistive index between zero point five five and zero point eight is normal. Here the adult and the paediatric data agree. In healthy children, Westra and colleagues found a resistive index of about zero point six in 1995. The index falls slowly with age.

The second rule says that loss of the triphasic wave in the hepatic veins means a stiff liver. This is an adult rule only. In 2000, Jequier and colleagues looked at one hundred healthy children. Only forty-two of them had a triphasic wave in all three hepatic veins. And not a single healthy newborn did. So a flat hepatic vein wave in a child is not, on its own, a sign of disease. Compare it with the child's earlier scans.

The third rule says that a pulsatile portal vein means right-heart disease, such as tricuspid regurgitation. In children with liver disease, the paediatric data contradict this rule. In the Westra study, in children with end-stage liver disease, a pulsatile portal vein had a sensitivity of ninety-four percent and a specificity of ninety percent for portal hypertension. So in a child with known liver disease, a pulsing portal vein should make you think of portal hypertension first, not of the heart.

The fourth rule says that a spleen longer than thirteen centimetres is enlarged. This is an adult rule only. In children, the upper limit is six centimetres at three months and nine centimetres at four years. We will come back to this.

The last three rules are about the liver graft. In adults, a hepatic artery resistive index under zero point five after transplant suggests stenosis. In children, the paediatric data differ. In uncomplicated paediatric grafts, the lowest five percent of values on the day of transplant were below zero point four three, and at one year below zero point five three. So a low index in the first days can be normal.

In adults, a portal vein peak velocity above one hundred and twenty-five centimetres per second suggests stenosis. In children, the paediatric data contradict this. The upper limit of normal was one hundred and seventy-one on the day of transplant, but only sixty at one year. So the adult threshold is too strict in the first week and much too lax later on.

And in adults, a hepatic artery peak velocity above two hundred centimetres per second suggests stenosis. In children, the upper limit of normal was between ninety-four and one hundred and forty-two at every time point up to two years. The adult threshold would never be reached in a normal child, and may not be reached in a child with a real problem.

These graft numbers come from Groningen, in the Netherlands, from a study by Verhagen and colleagues in 2023. We will look at it in detail in part eight.

Part six. Reversed portal flow and the hepatic artery as a warning sign.

Normally, blood in the portal vein flows into the liver. When the resistance inside the liver becomes very high, the flow can slow down, stop, or even reverse. Reversed, or hepatofugal, flow is a sign of advanced disease.

The classic study is from Gaiani and colleagues in 1991. Among two hundred and twenty-eight adults with cirrhosis, eight point three percent had spontaneous hepatofugal flow. It was more common in advanced cirrhosis: fifteen point four percent in Child class C, against two point seven percent in Child class A.

Interestingly, reversed flow is not all bad. It can decompress the varices. In the Gaiani study, none of the fifteen patients with reversed flow had a variceal bleed, against nine of twenty-nine with forward flow. But in adult studies, reversed flow comes with more encephalopathy, twenty-one percent against seven percent, because more portal blood bypasses the liver.

In children, reversed flow has a different meaning. In a series of one hundred and twenty-eight children with biliary atresia under three years of age, published by Gu and colleagues in 2015, forty-one percent had hepatofugal flow before transplant. And a small portal vein, four millimetres or less, predicted early

portal vein thrombosis after transplant, with a sensitivity of eighty-nine percent and a specificity of seventy-two percent. A small, slow, reversed vein is a vein that is likely to clot. So before a transplant in a child with atresia, the report must give the diameter and the direction of the portal vein. The surgeon plans the portal reconstruction from these two numbers.

Now the hepatic artery. The liver has two blood supplies, the portal vein and the hepatic artery. When the portal supply fails, one might expect the artery to open up. But in end-stage disease, the opposite happens downstream. The small vessels inside the liver resist the flow, and diastolic flow in the artery disappears. The resistive index rises.

In the Westra study, the resistive index was about zero point six in healthy children and in acute hepatitis, but zero point eight nine in children with end-stage liver disease waiting for a transplant.

Two European centres turned this into a prognostic tool for biliary atresia. At King's College Hospital in London, Broide and colleagues in 1997 studied sixty-four children with atresia. When the resistive index was one point zero or higher, meaning no diastolic flow at all, the two-year survival was fifty-two percent. When it was below one point zero, the two-year survival was ninety-four percent. King's used this to list children earlier for transplant, and repeated the Doppler every two to three months while the child waited.

In Hannover, Kadorff and colleagues in 1999 used three signs together: a portal vein speed below sixteen centimetres per second, a flat hepatic vein wave, and a hepatic artery resistive index of zero point eight or more. The Doppler study of the portal and hepatic veins predicted the two-year outcome with an accuracy of ninety-three percent.

These are useful numbers. But we should be honest about their limits. Neither threshold has been validated outside the centre that described it. And neither is part of the PELD score, the paediatric end-stage liver disease score that decides priority on the waiting list in many countries. So a high resistive index is a reason to talk to the transplant team early, not an automatic listing criterion.

Part seven. The spleen and portal hypertension.

The spleen is the simplest marker of portal hypertension. But its normal size changes a lot with age.

The standard table comes from Rosenberg and colleagues in 1991. They measured the spleen in two hundred and thirty children, from birth to twenty years, on a coronal view through the hilum, during quiet breathing. The upper limits of normal were about six centimetres at three months, six and a half at six months,

seven at one year, eight at two years, nine at four years, nine and a half at six years, ten at eight years, eleven at ten years, and eleven and a half at twelve years. After fifteen years, the limit was twelve centimetres in girls and thirteen in boys.

So a spleen of nine centimetres is normal in a ten-year-old and massive in a three-month-old.

There is one problem with age tables. Children with chronic liver disease are often short for their age. For them, a table based on height and weight is better. Megremis and colleagues published such a table in 2004, based on five hundred and twelve children. Even better is a z score, which tells you how many standard deviations the spleen is above the average for age.

Can the spleen, together with the platelet count, tell us who has varices? This was tested by Gana and colleagues in 2011. They studied one hundred and eight children with chronic liver disease or portal vein obstruction at eight centres, and every child had an endoscopy by someone who did not know the ultrasound results. They measured how well each test predicted any oesophageal varices, as the area under the curve. The ratio of the platelet count to the spleen z score was the best, at zero point eight four. A clinical prediction rule, combining platelets, spleen size and albumin, reached zero point eight zero. The platelet count alone reached zero point seven nine.

What does that mean in practice? The positive predictive value was about zero point eight six to zero point eight seven, and the negative predictive value was about zero point six three to zero point six four. But remember that sixty-nine percent of these children had varices. A negative predictive value of about two in three is not enough to avoid endoscopy. So these tools are good enough to decide who should have an endoscopy first. They are not good enough to decide who can skip it. And the spleen only works here as a z score, not as a raw length.

Beyond the spleen, the Doppler study can tell us what kind of portal hypertension a child has. There are three main patterns.

The first pattern is cirrhosis, where the block is inside the liver. The liver surface is nodular, and the caudate lobe is enlarged. There are collateral vessels, such as a reopened umbilical vein and shunts between the spleen and the kidney. The portal flow is slow, pulsatile or even reversed. In adults, the Baveno seven consensus of 2022 says that collaterals seen on imaging are enough to prove clinically significant portal hypertension in compensated advanced chronic liver disease, and the American society adopted this in 2024. This has not been tested in children.

The second pattern is extrahepatic portal vein obstruction. Here the portal vein outside the liver is blocked, usually by an old clot, and replaced by a tangle of small veins called a cavernoma. The liver itself is normal, but the spleen is big. This is the child's disease. It is one of the commonest causes of variceal bleeding in children. The key question for the radiologist is this: is the left portal vein, in a fold of the liver called the Rex recess, still open? If it is, the child may be cured by a bypass operation, called the Meso-Rex bypass, which connects the mesenteric vein to the left portal vein and restores portal flow into the liver. The Baveno six paediatric consensus of 2016 recommends offering this operation whenever the anatomy allows. So the first scan in a child with a cavernoma must look at the Rex recess. That one measurement decides between a curative operation and lifelong surveillance of varices.

The third pattern is a congenital portosystemic shunt. Here the portal blood bypasses the liver through an abnormal vessel, and drains into the inferior vena cava. On the Doppler study, there is little or no portal flow inside the liver. These children can look well for years. But in a large international study by Baiges and colleagues in 2020, twenty-eight percent had developed encephalopathy by the age of thirty. They can also develop liver nodules and adenomas, and pulmonary hypertension. So every child with a shunt needs an echocardiogram to look for pulmonary hypertension, and closure of the shunt should be considered.

Part eight. The liver graft.

After a liver transplant, Doppler ultrasound is the main tool for watching the blood vessels of the graft. The problem is the one we met in part five. The thresholds used to diagnose stenosis were worked out in adults, and they do not fit children.

The best paediatric data come from Groningen. In 2023, Verhagen and colleagues measured the Doppler values in one hundred and twenty-three paediatric liver transplants without vascular complications. They then described the upper limit of normal, the ninety-fifth percentile, on each day after transplant.

For the hepatic artery, the upper limit of the peak systolic velocity was one hundred and forty-two centimetres per second on the day of transplant, one hundred and thirty-seven on day one, one hundred and eighteen on day three, one hundred and eleven on day seven, one hundred and twenty-five at one month, eighty-six at three months, ninety-one at one year and ninety-four at two years.

For the portal vein, it was one hundred and seventy-one on the day of transplant, one hundred and forty-six on day one, one hundred and twenty on day three, one

hundred and five on day seven, one hundred and eleven at one month, eighty-two at three months, sixty at one year and fifty-three at two years.

Now compare these with the adult thresholds. For the hepatic artery, the adult threshold for stenosis is two hundred centimetres per second. A normal child never gets close to that. For the portal vein, the adult threshold is one hundred and twenty-five. A normal child goes above it in the first days after transplant, and then stays far below it for the rest of the first two years. So the adult threshold would call a normal portal vein abnormal in the first week, and would miss a narrowed vein a year later.

The practical lesson is simple. Read every graft Doppler study against the normal values for that post-operative day.

What is the Doppler study hunting for? The most feared complication is hepatic artery thrombosis. In the North American registry called SPLIT, Ebel and colleagues in 2020 looked at three thousand eight hundred and one children. Hepatic artery thrombosis occurred within ninety days in seven point four percent, which is about one child in fourteen. Half of these children lost their graft. Interestingly, children who received a technical variant graft, such as a split or a living donor graft, had fewer thromboses than those who received a whole liver, with an odds ratio of zero point five nine.

Before a thrombosis, there may be a narrowing, or stenosis. Downstream of a stenosis, the artery wave becomes slow and small. This is called the tardus parvus pattern. It is defined by a resistive index under zero point five and a slow rise in systole, with a systolic acceleration time of more than eighty milliseconds. But this pattern is only a screening sign. In a study of three hundred and sixty-one adult transplants in Seoul, published by Park and colleagues in 2011, the tardus parvus pattern had a positive predictive value of only thirty-eight percent for stenosis. When a low peak velocity, forty-eight centimetres per second or less, was added, the positive predictive value rose to eighty-eight percent. These data are from adults, but the lesson applies to children too. Tardus parvus is a reason to look further, not a diagnosis.

The portal vein can also narrow at the join, or anastomosis. In a paediatric study by Suzuki and colleagues in 2008, a diameter of the portal anastomosis under three point five millimetres had a sensitivity of one hundred percent and a specificity of ninety-two percent for stenosis. And in children with a left-lobe graft, a widened umbilical portion of the left portal vein, more than one point five centimetres, can be the only clue to a hidden stenosis just before it.

How often should the graft be scanned? Most centres do a Doppler study every day in the first week, and again whenever the liver enzymes rise. If the Doppler

study is abnormal and the liver tests agree, the next step is a CT angiogram or a formal angiogram.

Part nine. What the guidelines say.

How do the guidelines compare on ultrasound and Doppler of the liver? There are eight points worth knowing.

First, the first imaging test in a cholestatic infant. The guidelines are aligned, because there is only one document. The joint NASPGHAN and ESPGHAN guideline of 2017 recommends a fasting ultrasound, which reliably excludes a choledochal cyst and gallstones.

Second, a normal ultrasound in suspected atresia. Again aligned, in the same joint document. A normal ultrasound does not rule out non-syndromic biliary atresia.

Third, cut-offs for the signs of atresia. Here there are no data in the guidelines. The joint guideline lists the signs as suggestive, but gives no thresholds for the triangular cord, the gallbladder length or the hepatic artery. The numbers in this chapter come from individual studies.

Fourth, contrast-enhanced ultrasound for a liver lesion in a child. Here the drug labels are split. In the United States, the drug agency, the FDA, has approved the contrast agent Lumason for liver lesions in children. In Europe, the same agent, sold as SonoVue, is licensed for liver imaging in adults only. The European ultrasound federation, EFSUMB, in its paediatric position statement of 2017, supports its use in children off label. So the same test is on label in Boston and off label in Brussels.

Fifth, suspected portal vein thrombosis. Aligned. Both the American guidance of 2020 and the European guideline of 2016 recommend Doppler ultrasound as the first test.

Sixth, extrahepatic portal vein obstruction in a child. Aligned, because there is one international consensus. The Baveno six paediatric consensus recommends offering a Meso-Rex bypass if the anatomy allows.

Seventh, proving clinically significant portal hypertension from imaging. This is adult only. Baveno seven, and the American society in 2024, accept collaterals on imaging as proof in adults with compensated advanced chronic liver disease. There are no paediatric data.

Eighth, Doppler thresholds after transplant. Here there are no guideline data. No guideline gives paediatric cut-offs. The only paediatric reference ranges come from a single centre, Groningen, in 2023.

So the guidelines agree on the basics. Scan first, fast the baby, do not trust a normal scan, and use Doppler first for a clot. On almost everything with a number, they are silent, and we rely on single studies.

Part ten. Eight things to remember, and from question to finding.

Let us bring this together into eight pearls.

First, scan after a four-hour fast, except in the infant with hypopituitarism, who may become hypoglycaemic.

Second, the triangular cord has a specificity of about ninety-seven to one hundred percent, but a sensitivity anywhere from twenty-three to eighty-five percent. Act on a positive. Ignore a negative.

Third, the ghost triad is a gallbladder shorter than one point nine centimetres, without a bright lining, and with an irregular outline. And a normal gallbladder at six weeks can be a ghost at eight.

Fourth, subcapsular flow and a hepatic artery wider than one point five millimetres are sensitive Doppler signs. They are reasons to look harder.

Fifth, hepatofugal flow means advanced disease. It brings fewer variceal bleeds but more encephalopathy, and in a child with atresia waiting for a transplant, a small reversed vein predicts thrombosis after the operation.

Sixth, in biliary atresia, a hepatic artery resistive index of one or more was linked with a two-year survival of fifty-two percent. Discuss earlier listing.

Seventh, read the spleen only by age, height or z score. And use the ratio of platelets to the spleen z score to decide who needs an endoscopy first.

Eighth, after transplant, use paediatric ranges for each post-operative day. Tardus parvus is a screening sign, not a diagnosis.

To finish the teaching part of the chapter, here are the common clinical questions, and for each one, what to look for, which number matters, and what to do next.

Is it biliary atresia? Look for the ghost triad, the triangular cord, subcapsular flow and a large hepatic artery. The numbers are a cord over four millimetres, a gallbladder under one point nine centimetres, and an artery over one point five millimetres. If the scan is positive, go to the cholangiogram. If it is negative, it excludes nothing.

Is there a cyst at the hilum? Look at the size of the cyst, whether the ducts inside the liver are dilated, and the gallbladder. A small cyst with an abnormal

gallbladder suggests cystic atresia rather than a choledochal cyst. Both need surgery, but the timing differs, and cystic atresia is urgent.

Is there a splenic malformation? Look for several spleens, a preduodenal portal vein and an absent inferior vena cava. If present, ask for an echocardiogram and tell the surgeon.

Is there portal hypertension? Look at the spleen, the collaterals and the portal flow. Use the age limit for the spleen and the ratio of platelets to spleen z score, and use them to decide who needs an endoscopy first.

Is it extrahepatic portal vein obstruction? Look for a cavernoma and check the Rex recess. If the left portal vein is open, refer for a Meso-Rex bypass.

Is a child with atresia getting worse? Look at the hepatic artery resistive index, the portal vein speed and the hepatic vein wave. A resistive index of one or more, or a portal vein speed below sixteen centimetres per second, should lead to a discussion about earlier listing.

Is a child with atresia about to have a transplant? Report the diameter and direction of the portal vein. A vein of four millimetres or less, or reversed flow, means the surgeon must plan the portal reconstruction.

Is the graft in trouble? Look at the hepatic artery wave and the portal anastomosis. Compare with the ninety-fifth percentile for that post-operative day, and remember that a portal anastomosis under three point five millimetres suggests stenosis. If the Doppler study and the liver tests agree, go to a CT angiogram or a formal angiogram.

Four clinical cases.

Case one. A five-week-old girl has jaundice and pale stools. Her conjugated bilirubin and GGT are high. She was fed an hour before the scan. The ultrasound shows a small gallbladder with an irregular outline, and no triangular cord is seen. The report was done with a standard abdominal probe. The team is reassured by the absent cord. Do you agree? No. First, she was not fasted, so the small gallbladder may simply be contracted, and the scan should be repeated after a four-hour fast. Second, the triangular cord needs a high-frequency linear probe, and even then a negative cord changes nothing, because its sensitivity can be as low as twenty-three percent in infants of this age. Third, the guideline is clear that a normal ultrasound does not rule out atresia. With pale stools and a rising GGT, she needs a biopsy or a cholangiogram without delay. And look at the spleen and the inferior vena cava for the splenic malformation syndrome.

Case two. A seven-month-old boy with biliary atresia had a Kasai operation at nine weeks. He is still jaundiced. The Doppler study shows a hepatic artery with

no diastolic flow, a resistive index of one point zero, and a portal vein speed of twelve centimetres per second. His PELD score is not yet high. What does the Doppler study add? In the King's series, a resistive index of one or more was linked with a two-year survival of fifty-two percent, against ninety-four percent below that. The Hannover criteria, a portal vein speed under sixteen and a resistive index of zero point eight or more, point the same way. These thresholds are not validated outside those centres and are not part of PELD, but together they are a strong reason to discuss earlier assessment for transplant. Ask the radiologist to report the portal vein diameter and direction as well, because a small, reversed vein predicts thrombosis after transplant, and the surgeon needs to plan for it.

Case three. A six-year-old boy presents with a large variceal bleed. His liver tests are normal. His spleen is fourteen centimetres long. The ultrasound shows a tangle of small veins at the porta hepatis instead of a normal portal vein. What is the one question that the next scan must answer? This is extrahepatic portal vein obstruction with a cavernoma. His liver is normal, and his spleen is well above the upper limit of about nine and a half centimetres for his age. The key question is whether the left portal vein in the Rex recess is open. If it is, the Baveno six paediatric consensus recommends offering a Meso-Rex bypass, which can restore portal flow to the liver and cure the portal hypertension. So the scan, and if needed a CT or MR study, must look specifically at the Rex recess.

Case four. A two-year-old girl had a left lateral segment liver transplant four days ago. The Doppler report says that the hepatic artery peak velocity is one hundred and ten centimetres per second, with a resistive index of zero point four eight, and that the portal vein velocity is one hundred and thirty centimetres per second. The report calls the artery "possibly stenotic" because of the low resistive index, and the portal vein "possibly stenotic" because it is above one hundred and twenty-five. Her liver tests are improving. What do you think? Both conclusions use adult thresholds. In the Groningen data, a resistive index as low as zero point four three is still within the normal range on the day of transplant, and the upper limit of normal for portal vein velocity on day three was about one hundred and twenty, with higher values on the first two days. So these numbers are close to the normal range for this post-operative day. With improving liver tests, the right answer is to repeat the Doppler study the next day, look for a tardus parvus wave and measure the portal anastomosis, rather than to go straight to angiography. If the enzymes rise, or the artery wave becomes slow and small with a low peak velocity, the picture changes.

Key points to remember.

A complete report covers seven items: gallbladder, porta hepatis, hepatic artery, portal vein, hepatic veins, spleen, and situs with collaterals.

Scan the cholestatic infant after a four-hour fast, with a high-frequency linear probe. The exception is the infant with hypopituitarism.

The shape signs of atresia, the triangular cord and the ghost gallbladder, are specific. The Doppler signs, subcapsular flow and a large hepatic artery, are sensitive.

The triangular cord has a specificity of about ninety-seven to one hundred percent, but a sensitivity between twenty-three and eighty-five percent. A negative cord changes nothing.

The ghost triad is a gallbladder under one point nine centimetres, without a bright lining, with an irregular outline. The gallbladder can lag behind the disease.

A normal ultrasound does not rule out biliary atresia. Pale stools or a rising GGT mean biopsy or cholangiogram.

Many healthy children do not have a triphasic hepatic vein wave. In a child with liver disease, a pulsatile portal vein suggests portal hypertension.

Hepatofugal flow means advanced disease. Before transplant in atresia, report the portal vein diameter and direction.

A hepatic artery resistive index of one or more in atresia was linked with poor two-year survival. It is a reason to talk to the transplant team early.

Read spleen size by age, or better by z score. The ratio of platelets to spleen z score helps decide who needs endoscopy first, but cannot replace it.

In extrahepatic portal vein obstruction, check the Rex recess on the first scan. A patent left portal vein means a Meso-Rex bypass may be possible.

After transplant, adult Doppler thresholds do not fit. Use paediatric ranges for each post-operative day.

Hepatic artery thrombosis occurs in about seven percent of paediatric grafts. Tardus parvus is a screening sign, not a diagnosis.

Questions to test yourself.

Question one. A cholestatic infant has a normal ultrasound, but the stools remain pale. What does the guideline say? Answer: a normal ultrasound does not rule out

biliary atresia. The next step is a biopsy or a cholangiogram, and the scan can be repeated, because the gallbladder can lag behind the disease.

Question two. What are the three parts of the gallbladder ghost triad? Answer: a length under one point nine centimetres, no smooth bright lining, and an irregular or lobulated outline.

Question three. Why is the sensitivity of the triangular cord so variable between studies? Answer: because of different cut-offs, three or four millimetres, because the fibrous plate may not have formed in young infants, and because it needs a high-frequency linear probe.

Question four. A healthy newborn has a flat hepatic vein wave on Doppler. Is this abnormal? Answer: no. In the Jequier study, no healthy newborn had a triphasic wave, and only forty-two of one hundred healthy children had it in all three veins.

Question five. In a child with end-stage liver disease, the portal vein flow pulses with the heart. What does this suggest? Answer: portal hypertension. In the Westra study, a pulsatile portal vein had a sensitivity of ninety-four percent and a specificity of ninety percent for portal hypertension in these children.

Question six. What was the two-year survival in the King's series of children with atresia whose hepatic artery resistive index was one or more? Answer: fifty-two percent, against ninety-four percent when it was below one.

Question seven. In a child with a portal cavernoma, what is the most important structure to look for on ultrasound, and why? Answer: the left portal vein in the Rex recess. If it is open, the child may be offered a Meso-Rex bypass, which can cure the portal hypertension.

Question eight. Why should adult Doppler thresholds not be used after a paediatric liver transplant? Answer: because normal values in children change with each post-operative day. The normal hepatic artery never reaches the adult threshold of two hundred centimetres per second, and the normal portal vein exceeds the adult threshold of one hundred and twenty-five in the first days, then falls far below it.

End of chapter.